# CHEMISTRY

## SECTION - A

**Multiple Choice Questions:** This section contains 20 multiple choice questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

**Choose the correct answer :**

61. Match List I with List II.

| | List I (Substances) | | List II (Element Present) |
|---|---|---|---|
| A. | Ziegler catalyst | I. | Rhodium |
| B. | Blood pigment | II. | Cobalt |
| C. | Wilkinson catalyst | III. | Iron |
| D. | Vitamin $B_{12}$ | IV. | Titanium |

Choose the correct answer from the options given below.

(1) A-II, B-IV, C-I, D-III (2) A-III, B-II, C-IV, D-I

(3) A-II, B-III, C-IV, D-I (4) A-IV, B-III, C-I, D-II

**Answer (4)**

**Sol.** Ziegler catalyst – $TiCl_4 \cdot Al(C_2H_5)_3$

Blood pigment – Compound containing iron

Wilkinson catalyst – $[RhCl(PPh_3)_3]$

Vitamin $B_{12}$ – Compound containing cobalt

$\therefore$ Correct match is A-IV, B-III, C-I, D-II.

62. Given below are two statements :

**Statement I :** The electronegativity of group 14-elements from Si to Pb, gradually decreases.

**Statement II :** Group 14 contains non-metallic, metallic, as well as metalloid elements.

In the light of the above statements, choose the **most appropriate** from the options given below.

(1) Both Statement I and Statement II are true

(2) Both Statement I and Statement II are false

(3) Statement I is true but Statement II is false

(4) Statement I is false but Statement II is true

**Answer (4)**

**Sol.** Statement-I is false because electronegativity of Group-14 elements from Si to Pb is almost constant, *i.e.*, 1.8.

Statement-II is correct because Group-14 contains non-metals, metalloids as well as metals.

| | | |
|---|---|---|
| Carbon | – | Non-metal |
| Silicon | – | Metalloid |
| Germanium | – | Metalloid |
| Tin | – | Metal |
| Lead | – | Metal |

63. In chromyl chloride test for confirmation of $Cl^-$ ion, a yellow solution is obtained. Acidification of the solution and addition of amyl alcohol and 10% $H_2O_2$ turns organic layer blue indicating formation of chromium pentoxide. The oxidation state of chromium in that is

(1) +6 (2) +10

(3) +5 (4) +3

**Answer (1)**

**Sol.** The reactions involved in chromyl chloride test and subsequent formation of chromium pentaoxide are as follows :

$$Cr_2O_7^{2-} + 6H^+ + 4Cl^- \longrightarrow \underset{\text{(Red)}}{2CrO_2Cl_2(g)} + 3H_2O$$

$$CrO_2Cl_2(g) + 4NaOH(aq) \longrightarrow \underset{\text{(Yellow)}}{Na_2CrO_4(aq)} + 2NaCl + 2H_2O$$

$$\underset{\text{(Yellow)}}{2Na_2CrO_4(aq)} + 2H^+ \longrightarrow \underset{\text{(Orange)}}{Na_2Cr_2O_7(aq)} + 2Na^+ + H_2O$$

$$Na_2Cr_2O_7 + 4H_2O_2 + 2H^+ \longrightarrow \underset{\substack{\text{(Blue in}\\ \text{amyl alcohol)}}}{2CrO_5} + 5H_2O + 2Na^+$$

Oxidation state of chromium in $CrO_5$ is +6.



64. Identify the incorrect pair from the following.

(1) Cryolite – $Na_3AlF_6$

(2) Fluoroapatite – $3Ca_3(PO_4)_2 \cdot CaF_2$

(3) Fluorspar – $BF_3$

(4) Carnallite – $KCl \cdot MgCl_2 \cdot 6H_2O$

**Answer (3)**

**Sol.** The correct pairs are

1. Cryolite – $Na_3AlF_6$
2. Fluorapatite – $3Ca_3(PO_4)_2 \cdot CaF_2$
3. Fluorspar – $CaF_2$
4. Carnallite – $KCl \cdot MgCl_2 \cdot 6H_2O$

65. Given below are two statements : one is labelled as **Assertion A** and the other is labelled as **Reason R**.

**Assertion A :** Aryl halides cannot be prepared by replacement of hydroxyl group of phenol by halogen atom.

**Reason R :** Phenols react with halogen acids violently.

In the light of the above statements, choose the **most appropriate** from the options given below.

(1) **A** is false but **R** is true

(2) Both **A** and **R** are true but **R** is NOT the correct explanation of **A**

(3) **A** is true but **R** is false

(4) Both **A** and **R** are true and **R** is the correct explanation of **A**

**Answer (3)**

**Sol.** Assertion is true because aryl halides cannot be prepared by replacement of OH group of phenol by halogen atom.

Reason is false because phenols do not react with halogen acids.

66. Type of amino acids obtained by hydrolysis of proteins is :

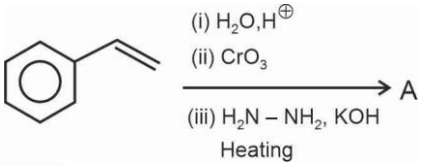
(1) $\beta$ (2) $\gamma$

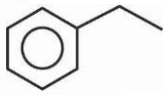
(3) $\delta$ (4) $\alpha$

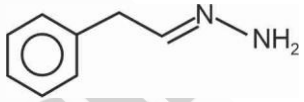
**Answer (4)**

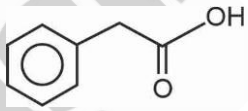
**Sol.** Hydrolysis of proteins results in the formation of $\alpha$-amino acids only.

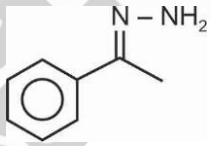
67. The final product A formed in the following multistep reaction sequence is




(1) 

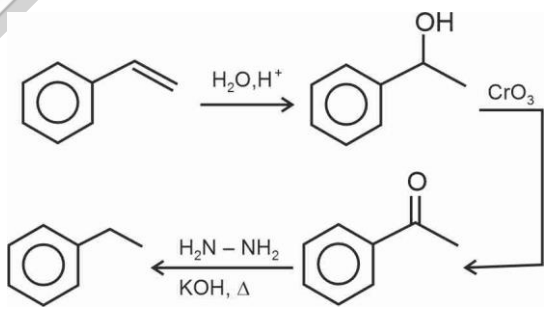
(2) 


(3) 


(4) 


**Answer (1)**

**Sol.**






68. Appearance of blood red colour, on treatment of the sodium fusion extract of an organic compound with $FeSO_4$ in presence of concentrated $H_2SO_4$ indicates the presence of element/s

(1) N and S (2) N

(3) Br (4) S

**Answer (1)**

**Sol.** Appearance of blood red colour on treatment of sodium fusion extract of an organic compound with $FeSO_4$ in presence of conc. $H_2SO_4$ is due to the formation of $[Fe(SCN)]^{2+}$

$Na + C + N + S \rightarrow NaSCN$

$Fe^{3+} + SCN^- \rightarrow [Fe(SCN)]^{2+}$ (Blood red)

This indicates the presence of both N and S in the organic compound.

69. The difference in energy between the actual structure and the lowest energy resonance structure for the given compound is

(1) hyperconjugation energy

(2) electromeric energy

(3) resonance energy

(4) ionization energy

**Answer (3)**

**Sol.** The difference in energy between the actual structure and the lowest energy resonance structure of the given compound is resonance energy.

70. In alkaline medium, $MnO_4^-$, oxidises $I^-$ to

(1) $IO^-$ (2) $I_2$

(3) $IO_3^-$ (4) $IO_4^-$

**Answer (3)**

**Sol.** In alkaline medium, $MnO_4^-$ oxidises $I^-$ to $IO_3^-$.

$$2MnO_4^- + I^- + H_2O \longrightarrow 2MnO_2 + IO_3^- + 2OH^-$$

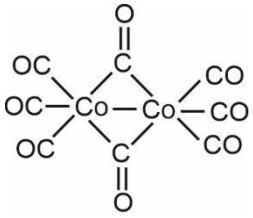
71. In which one of the following metal carbonyls, CO forms a bridge between metal atoms?

(1) $[Mn_2(CO)_{10}]$ (2) $[Co_2(CO)_8]$

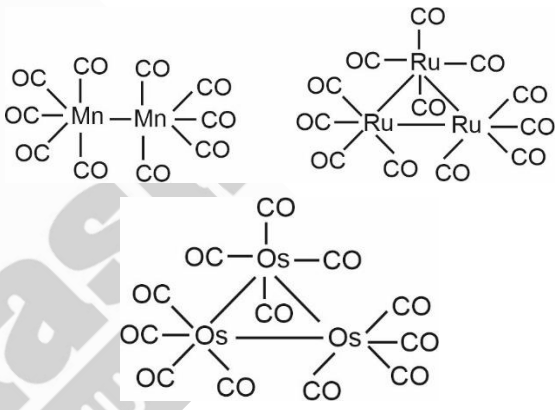
(3) $[Ru_3(CO)_{12}]$ (4) $[Os_3(CO)_{12}]$

**Answer (2)**

**Sol.** Among the given metal carbonyls, only $[Co_2(CO)_8]$ shows bridging CO ligands between two metal atoms.



Structures of other metal carbonyls are



72. Chlorine undergoes disproportionation in alkaline medium as shown below :

$$aCl_{2(g)} + bOH^-_{(aq)} \rightarrow cClO^-_{(aq)} + dCl^-_{(aq)} + eH_2O_{(l)}$$

The values of a, b, c and d in a balanced redox reaction are respectively :

(1) 2, 4, 1 and 3 (2) 1, 2, 1 and 1

(3) 2, 2, 1 and 3 (4) 3, 4, 4 and 2

**Answer (2)**

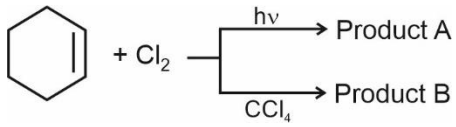
**Sol.** The balanced disproportionation reaction of $Cl_2$ in alkaline medium is

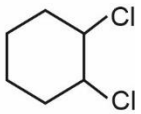
$Cl_2 + 2OH^- \rightarrow ClO^- + Cl^- + H_2O$

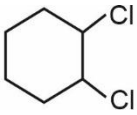
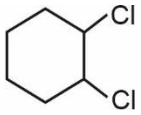


$\therefore$ a = 1, b = 2, c = 1, d = 1

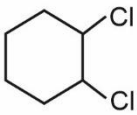
73. Identify product A and product B :



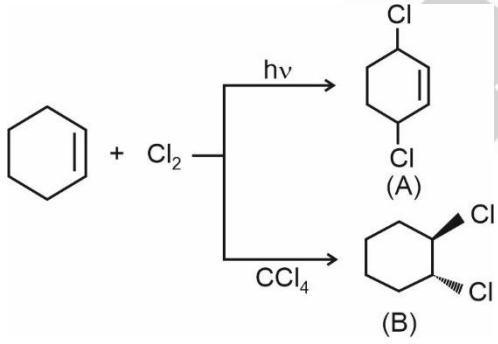

(1) A :  B : 

(2) A :  B : 

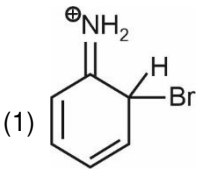
(3) A :  B : 

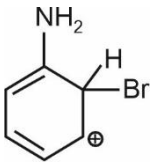
(4) A :  B : 

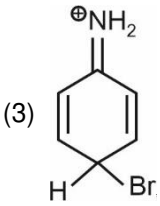
**Answer (1)**

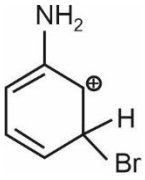
**Sol.** 


74. The arenium ion which is not involved in the bromination of Aniline is _____.

(1) 

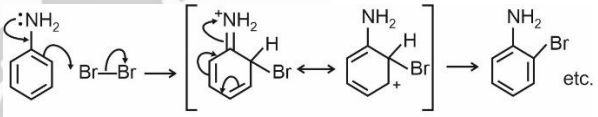

(2) 

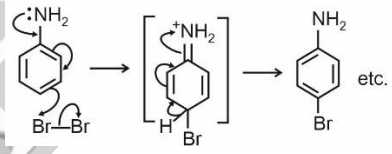

(3) 


(4) 


**Answer (4)**

**Sol.** Bromination of aniline results in the formation of 2, 4, 6-tribromoaniline because $NH_2$ group strongly activates the benzene ring particularly at the ortho and para positions. So, arenium ion will be formed when $Br_2$ attacks the ortho and para positions but not the meta position. Therefore arenium ion having Br atom at the meta position will not be involved in the given reaction.




75. Given below are two statements : one is labelled as **Assertion A** and the other is labelled as **Reason R**

**Assertion A:** The first ionisation enthalpy decreases across a period.

**Reason R:** The increasing nuclear charge outweighs the shielding across the period.

In the light of the above statements, choose the ***most appropriate*** from the options given below :



(1) **A** is true but **R** is false

(2) **A** is false but **R** is true

(3) Both **A** and **R** are true but **R** is NOT the correct explanation of **A**

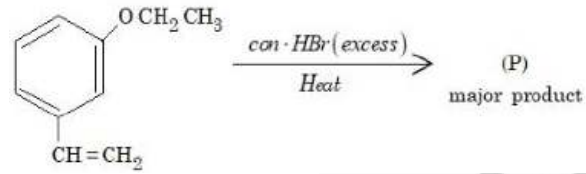
(4) Both **A** and **R** are true and **R** is the correct explanation of **A**

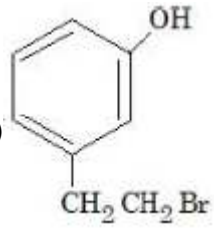
**Answer (2)**

**Sol.** Assertion is false because the first ionisation enthalpy generally increases across a period barring few exceptions.

Reason is true because the increase in first ionisation enthalpy across a period is due to the increasing nuclear charge outweighs the shielding effect.

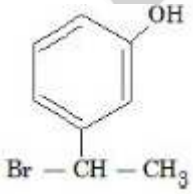
76. The major product(P) in the following reaction is




(1) 

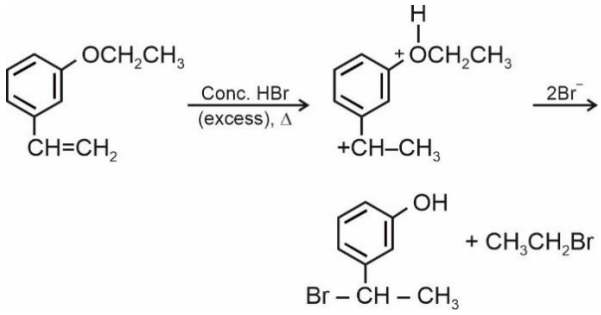

(2) 


(3) 


(4) 


**Answer (4)**

**Sol.**




77. Which of the following is **not** correct?

(1) $\Delta G$ is negative for a spontaneous reaction

(2) $\Delta G$ is positive for a non-spontaneous reaction

(3) $\Delta G$ is zero for a reversible reaction

(4) $\Delta G$ is positive for a spontaneous reaction

**Answer (4)**

**Sol.** $\Delta G$ is negative for a spontaneous process, zero for a reversible process and positive for a non- spontaneous process.

78. The correct set of four quantum numbers for the valence electron of rubidium atom (Z = 37) is:

(1) $5, 0, 0, +\frac{1}{2}$ (2) $5, 0, 1, +\frac{1}{2}$

(3) $5, 1, 0, +\frac{1}{2}$ (4) $5, 1, 1, +\frac{1}{2}$

**Answer (1)**

**Sol.** The electronic configuration of Rb-atom (Z = 37) is

$1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^6 5s^1$

The values of four quantum numbers of the valence electron of Rb-atom are

$n = 5, l = 0, m = 0, s = +\frac{1}{2}$

79. $KMnO_4$ decomposes on heating at 513K to form $O_2$ along with

(1) $K_2MnO_4$ & $MnO_2$ (2) $K_2MnO_4$ & Mn

(3) $MnO_2$ & $K_2O_2$ (4) Mn & $KO_2$

**Answer (1)**



**Sol.** $KMnO_4$ decomposes on heating at 513 K as per the following reaction

$$2KMnO_4 \xrightarrow{513\ K} K_2MnO_4 + MnO_2 + O_2$$

Therefore, $K_2MnO_4$ and $MnO_2$ are formed alongwith $O_2$.

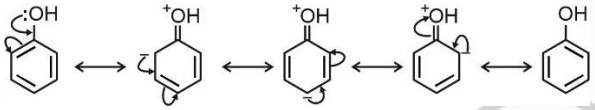
80. The interaction between $\pi$ bond and lone pair of electrons present on an adjacent atom is responsible for

(1) Hyperconjugation (2) Inductive effect

(3) Resonance effect (4) Electromeric effect

**Answer (3)**

**Sol.** The interaction between $\pi$ bond and lone pair of electrons present on adjacent atom is responsible for 'Resonance effect'.



## SECTION - B

**Numerical Value Type Questions:** This section contains 10 Numerical based questions. Attempt any 5 questions out of 10. The answer to each question should be rounded-off to the nearest integer.

81. The osmotic pressure of a dilute solution is $7 \times 10^5$ Pa at 273K. Osmotic pressure of the same solution at 283K is _____ $\times 10^4$ $Nm^{-2}$.

**Answer (73)**

**Sol.** $P_1$, osmotic pressure at 273K = $7 \times 10^5$ Pa

$P_2$, osmotic pressure at 283K is given by

$$\frac{P_2}{P_1} = \frac{T_2}{T_1} \qquad (\because\ P \propto T)$$

$$P_2 = 7\times10^5 \times \frac{283}{273} = 72.56\times10^4 \text{ Pa}$$

$$\simeq 73 \times 10^4\ Nm^{-2}$$

82. The mass of zinc produced by the electrolysis of zinc sulphate solution with a steady current of 0.015 A for 15 minutes is _____ $\times 10^{-4}$ g.

(Atomic mass of zinc = 65.4 amu)

**Answer (46)**

**Sol.** I = 0.015 amp

t = 15 × 60 = 900 sec

Q = I × t

= 0.015 × 900 C = 13.5 C

$$= \frac{13.5}{96500} F$$

$Zn^{2+} + 2e \longrightarrow Zn$

$$\text{Number of moles of Zn produced} = \frac{13.5}{2\times96500}$$

$$\text{Mass of Zn produced} = \frac{13.5\times65.4}{2\times96500} g$$

$$= 45.75 \times 10^{-4}\ g$$

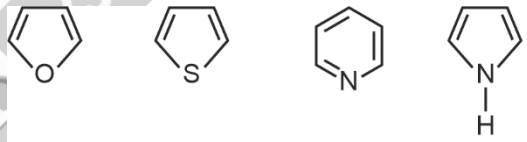
$$\simeq 46 \times 10^{-4}\ g$$

83. Number of compounds among the following which contain sulphur as heteroatom is _____.

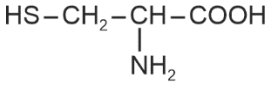
Furan, Thiophene, Pyridine, Pyrrole, Cysteine, Tyrosine

**Answer (2)**

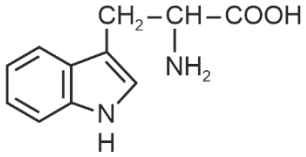
**Sol.** The structures of the given compounds are



Furan Thiophene Pyridine Pyrrole



Cysteine



Tyrosine

Number of compounds containing S-atom = 2



84. From the compounds given below, number of compounds which give positive Fehling's test is _____.

Benzaldehyde, Acetaldehyde, Acetone, Acetophenone, Methanal, 4-nitrobenzaldehyde, cyclohexane carbaldehyde.

**Answer (3)**

**Sol.** Fehling's test is given by aliphatic aldehydes only. The following compounds give positive Fehling's test.

Acetaldehyde, Methanal, Cyclohexane carbaldehyde

85. For a reaction taking place in three steps at same temperature, overall rate constant $K = \frac{K_1K_2}{K_3}$. If $Ea_1$, $Ea_2$ and $Ea_3$ are 40, 50 and 60 kJ/mol respectively, the overall Ea is _____ kJ/mol.

**Answer (30)**

**Sol.** The rate constant of a three steps reaction is

$$K = \frac{K_1K_2}{K_3}$$

If $Ea_1$, $Ea_2$ and $Ea_3$ are activation energies of three steps respectively then

$Ea_1 = 40\ kJ\ mol^{-1}$, $Ea_2 = 50\ kJ\ mol^{-1}$, $Ea_3 = 60\ kJ\ mol^{-1}$

Using Arrhenius equation,

$K = Ae^{-Ea/RT}$

$Ea = Ea_1 + Ea_2 - Ea_3$

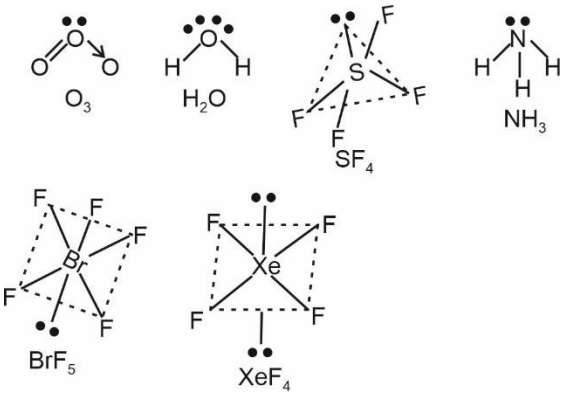
$= 40 + 50 - 60 = 30\ kJ\ mol^{-1}$

86. Number of compounds with one lone pair of electrons on central atom amongst following is _____

$O_3$, $H_2O$, $SF_4$, $ClF_3$, $NH_3$, $BrF_5$, $XeF_4$

**Answer (4)**

**Sol.** The structures of the given compounds are




$\therefore$ No. of compounds with one lone pair of electrons on central atom = 4

87. For the reaction $N_2O_{4(g)} \rightleftarrows 2NO_{2(g)}$, $K_p = 0.492$ atm at 300K. $K_C$ for the reaction at same temperature is _____ $\times 10^{-2}$.

(Given : R = 0.082 L atm $mol^{-1}$ $K^{-1}$)

**Answer (2)**

**Sol.** $N_2O_4(g) \rightleftharpoons 2NO_2(g)$ $K_p = 0.492$ atm at 300 K

$K_p = K_c(RT)^{\Delta n_g}$

$0.492 = K_c(0.082 \times 300)$ $(\because \Delta n_g = 1)$

$K_c = \frac{0.492}{24.6} = 2 \times 10^{-2}$

88. A solution of $H_2SO_4$ is 31.4% $H_2SO_4$ by mass and has a density of 1.25g/mL.

The molarity of the $H_2SO_4$ solution is _____M (nearest integer)

[Given molar mass of $H_2SO_4$ = 98g $mol^{-1}$]

**Answer (4)**

**Sol.** Density of $H_2SO_4$ solution = 1.25 $gmol^{-1}$

Mass percentage of $H_2SO_4$ = 31.4 %

Mass of 1 L $H_2SO_4$ solution = 1250 g

Mass of $H_2SO_4$ in 1L solution = $\frac{1250 \times 31.4}{100}$ g



Molarity of $H_2SO_4$ solution $= \frac{1250 \times 31.4}{98 \times 100}$

$= 4.00$ M

89. The number of species from the following which are paramagnetic and with bond order equal to one is _____.

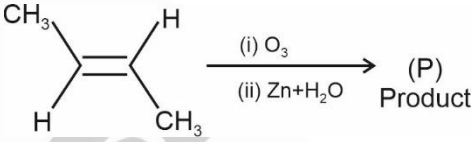
$H_2, He_2^+, O_2^+, N_2^{2-}, O_2^{2-}, F_2, Ne_2^+, B_2$

**Answer (1)**

**Sol.** According to M.O. theory electronic configurations of the given species are

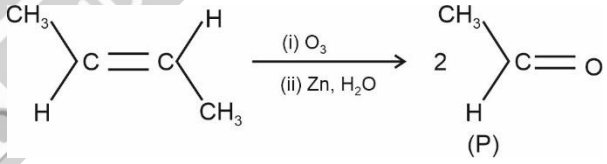
| | | | Bond order | |
|---|---|---|---|---|
| $H_2$ | : | $\sigma_{1s}^2$ | 1 | Diamagnetic |
| $He_2^+$ | : | $\sigma_{1s}^2\sigma^{*1}_{1s}$ | 0.5 | Paramagnetic |
| $O_2^+$ | : | $\sigma_{1s}^2\sigma^{*2}_{1s}\sigma_{2s}^2$ $\sigma^{*2}_{2s}\sigma_{2p_z}^2\pi_{2p_x}^2$ $\pi_{2p_y}^2\pi^{*1}_{2p_x}$ | 2.5 | Paramagnetic |
| $N_2^{2-}$ | : | $\sigma_{1s}^2\sigma^{*2}_{1s}\sigma_{2s}^2$ $\sigma^{*2}_{2s}\sigma_{2p_z}^2\pi_{2p_x}^2$ $\pi_{2p_y}^2\pi^{*1}_{2p_x}\pi^{*1}_{2p_y}$ | 2.0 | Paramagnetic |
| $F_2, O_2^{2-}$ | : | $\sigma_{1s}^2\sigma^{*2}_{1s}\sigma_{2s}^2$ $\sigma^{*2}_{2s}\sigma_{2p_z}^2\pi_{2p_x}^2$ $\pi_{2p_y}^2\pi^{*2}_{2p_x}$ $\pi^{*2}_{2p_y}$ | 1.0 | Diamagnetic |
| $Ne_2^+$ | : | $\sigma_{1s}^2\sigma^{*2}_{1s}\sigma_{2s}^2$ $\sigma^{*2}_{2s}\sigma_{2p_z}^2\pi_{2p_x}^2$ $\pi_{2p_y}^2\pi^{*2}_{2p_x}$ $\pi^{*2}_{2p_y}\sigma^{*1}_{2p_z}$ | 0.5 | Paramagnetic |
| $B_2$ | : | $\sigma_{1s}^2\sigma^{*2}_{1s}\sigma_{2s}^2$ $\sigma^{*2}_{2s}\pi_{2p_x}^1\pi_{2p_y}^1$ | 1.0 | Paramagnetic |

No. of compounds which are paramagnetic and with bond order equal to one is 1

90. 


Consider the given reaction. The total number of oxygen atom/s present per molecule of the product (P) is _____.

**Answer (1)**

**Sol.** 


No. of O-atom present per molecule of (P) = 1

❑ ❑ ❑

